



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	BE	Semester:	VIII
Course Code:		Course Name:	AAI

Name of Student:	BARI ANKIT VINOD
Roll No. :	61
Assignment No.:	5
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	03
Demonstrated Knowledge	3	02
Legibility	2	02
Total	10	07

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : Dr. Valshehi Shirsat

Signature :

Date :

6/14/26

Q.1) Compare bagging and boosting in terms of training approach, handling of bias and variance, sensitivity to noise and parallelization.

→ (i) Training Approach

- Bagging (bootstrap aggregation) -
models are trained independently using different bootstrap samples.
eg. Random forest.
- Boosting -
models are trained sequentially, where each new model focuses on correcting errors of previous models.

(ii) Handling bias and variance

- Bagging -
mainly reduces variance
works well for high variance models like decision tree.
does not significantly reduce bias.
- Boosting
reduces both bias and variance
converts weak learners into strong learners.

(iii) Sensitivity to noise

- Bagging -
less sensitive to noise.
since models are independent noise does not dominate learning.
- Boosting -
highly sensitive to noise.
misclassified noisy points get more importance
- Bagging - variance reduction, stable parallel
- Boosting - bias + variance reduction, powerful but sensitive.

Q. 2) Apply the XGBoost algorithm to the following dataset and the step-by-step calculation using 2 boosting iterations and squared error loss

given, $x : [1, 2, 3, 4]$

$y : [2, 3, 6, 8]$

→ S1. Initial prediction -

start with mean of y .

$$F_0 = \frac{2 + 3 + 6 + 8}{4} = 4.75$$

$$\text{Predictions } \hat{y} = [4.75, 4.75, 4.75, 4.75]$$

S2. Compute residuals -

$$r = [-2.75, -1.75, 1.25, 3.25]$$

S3. fit tree on residuals -

$$\text{left } (x \leq 2) : \text{mean} = (-2.75 + -1.75) / 2 = -2.25$$

$$\text{right } (x > 2) : \text{mean} = (1.25 + 3.25) / 2 = 2.25$$

S4. Update predictions

$$\epsilon = F_1 = F_0 = 0.5 = h$$

$$F_1 = [4.75 - 1.125, 4.75 - 1.125, 4.75 + 1.125, 4.75 + 1.125]$$

$$F_1 = [3.625, 3.625, 5.875, 5.875]$$

S5. Residuals -

$$r = [2 - 3.625, 3 - 3.625, 6 - 5.875, 8 - 5.875]$$

$$r = [-1.625, -0.625, 0.125, 2.125]$$

S6. Output

$$F_2 = [3.0625, 3.0625, 6.4375, 6.4375]$$

3) Compare decision tree and random forest algorithms. which model you would choose for a dataset prone to overfitting and justify your answer.

→ (i) Basic idea -

- decision tree - single model tree
 - splits data based on features.
- Random forest - ensemble of multiple decision trees using bagging.

(ii) Overfitting -

- decision tree - highly prone to overfitting
 - learns noise in data.
- Random forest - Reduces overfitting using averaging.
 - more robust and generalized.

(iii) Variance -

- decision tree - high variance
- Random forest - reduced variance

(iv) Interpretability -

- decision tree - easy to interpret.
- random forest - harder (many trees)

(v) Accuracy -

- decision tree - lower
- Random forest - Higher

- which model to choose?

- for dataset prone to overfitting, choose "Random forest", because uses multiple trees, reduces variance, uses random sampling feature selection, prevents model from memorizing noise.

Q. 4) Compare Adaboost and gradient boosting algorithms, which algorithm performs better in the presence of noisy data and justify your answer.

→ (i) working principle -

- AdaBoost - Adjust weights of data points.
 - misclassified samples get higher weight.
- Gradient boosting - minimizes loss using gradient descent
 - fits model to residual errors.

(ii) Handling errors -

- AdaBoost - focuses on misclassified points.
- Gradient Boosting - focuses on overall error minimization.

(iii) Sensitivity to Noise -

- AdaBoost - very sensitive to noise.
 - noisy points get high weight to overfitting.
- Gradient Boosting - less sensitive
 - can control overfitting using.

(iv) flexibility -

- AdaBoost - limited flexibility.
- Gradient Boosting - highly flexible.

(v) Performance -

- AdaBoost - good for clean dataset.
- Gradient Boosting - better overall performance.

which is better for noisy data?

Gradient boosting, does not over focused on noisy samples, uses gradient optimization instead of hard weighting.